MT1011 Engineering Materials

Strengthening mechanisms

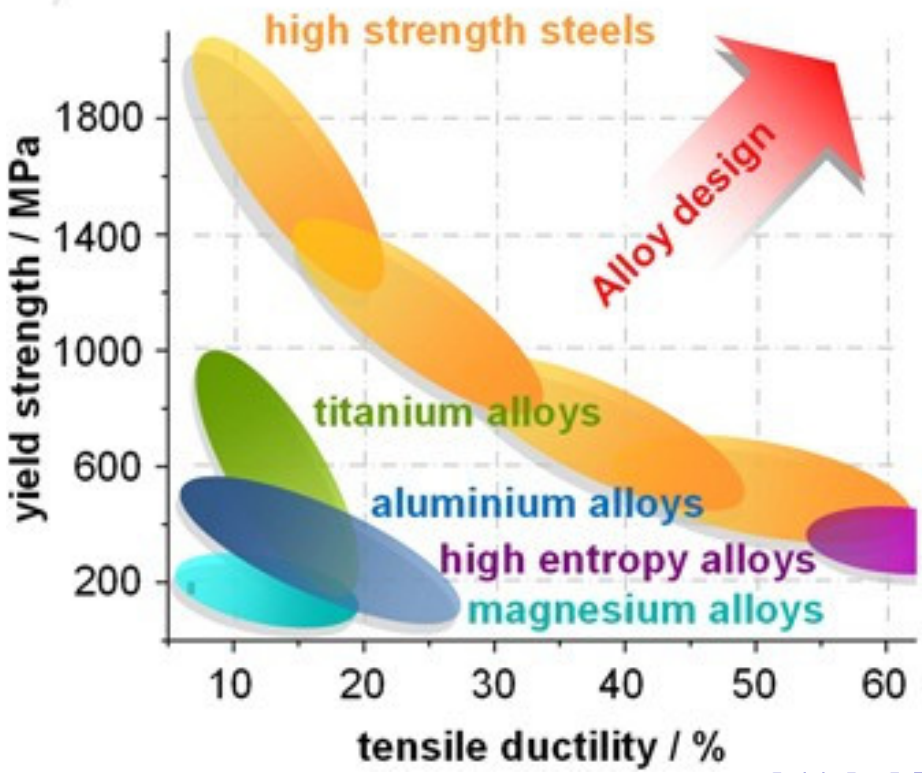
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October 19, 2026

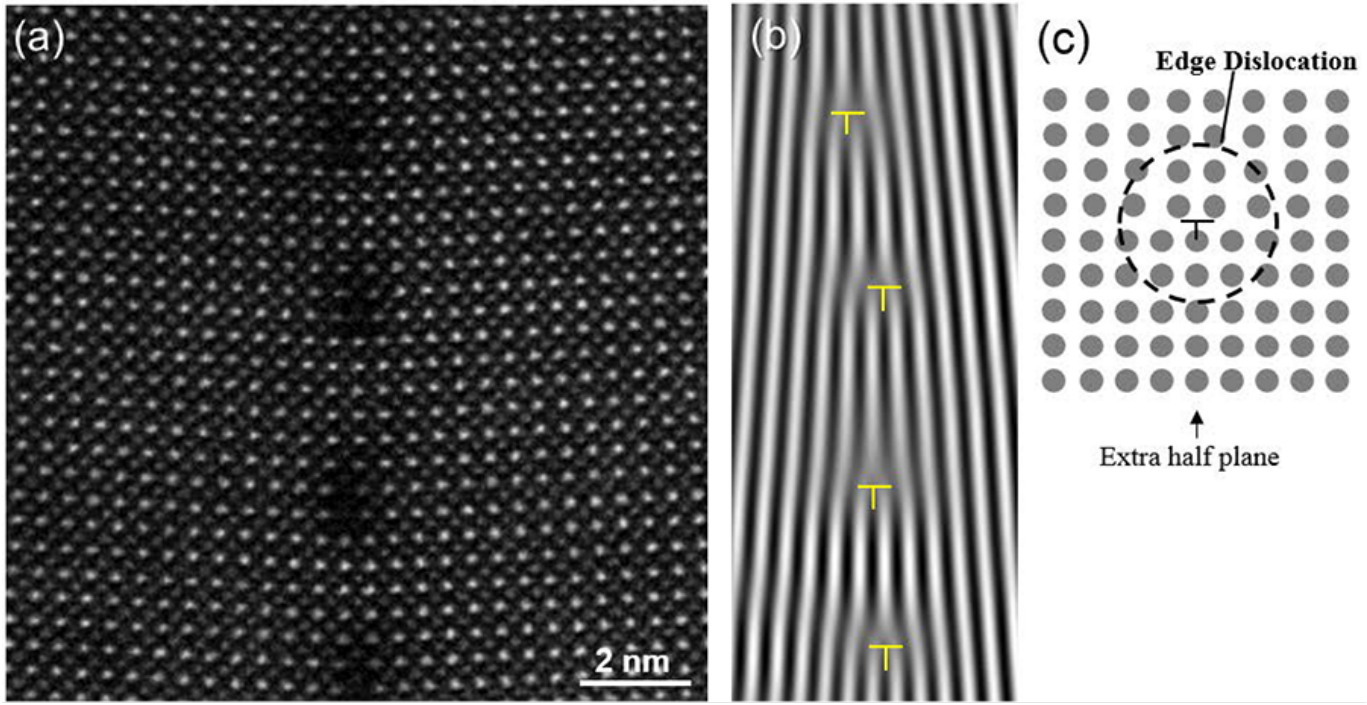
Strength and ductility

- Materials engineers need to design alloys with high strengths for various applications.
- Engineers sometimes have to change the mechanical properties of an existing alloy to suit an application.
- These materials should also have ductility and toughness.
- Normally, when an alloy is strengthened ductility is reduced.



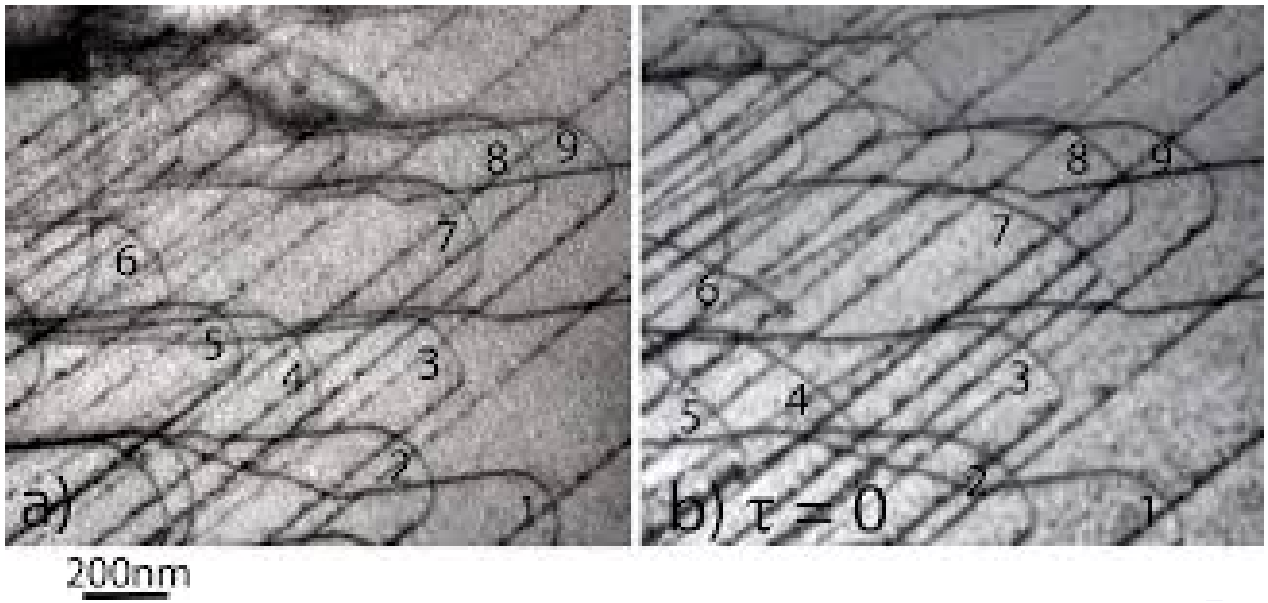
Strengthening

- Dislocation motion affects mechanical behaviour of metals.
- Plastic deformation observed is due to the motion of large numbers of dislocations.
- The ability of a metal to plastically deform depends on how easy it is to move the dislocations.



Strengthening

- Hardness and strength (yield and tensile) are related to how easy the plastic deformation can occur.
- By reducing dislocation mobility, strength can be increased.
- Greater mechanical forces will be required to start plastic deformation (high yield point).
- If the dislocation motion is easy, the metal can deform easily-the metal will be softer and weaker.
- All strengthening methods depend on making dislocation motion difficult (Below: Dislocation motion in Zr).



Dislocation motion

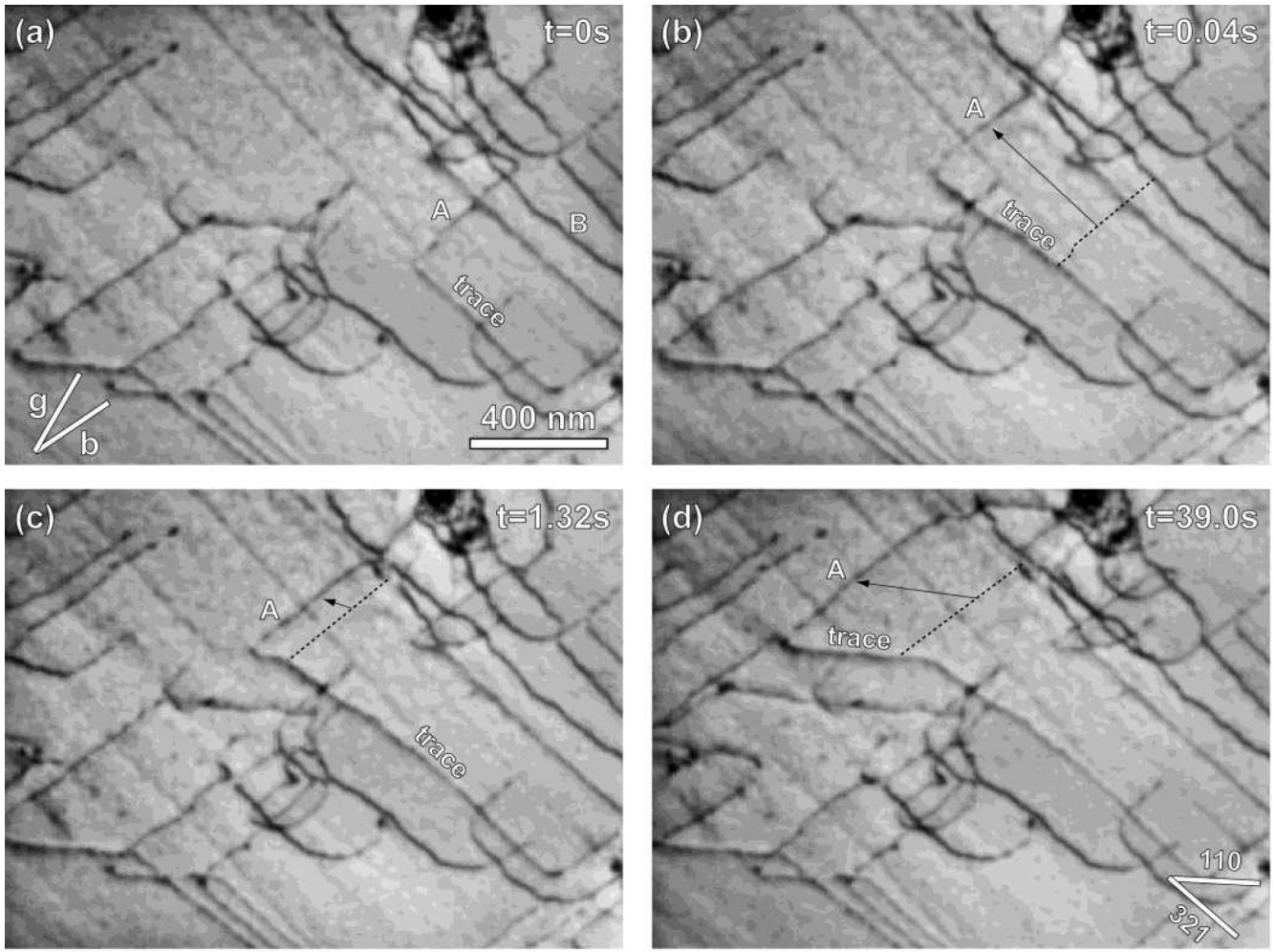
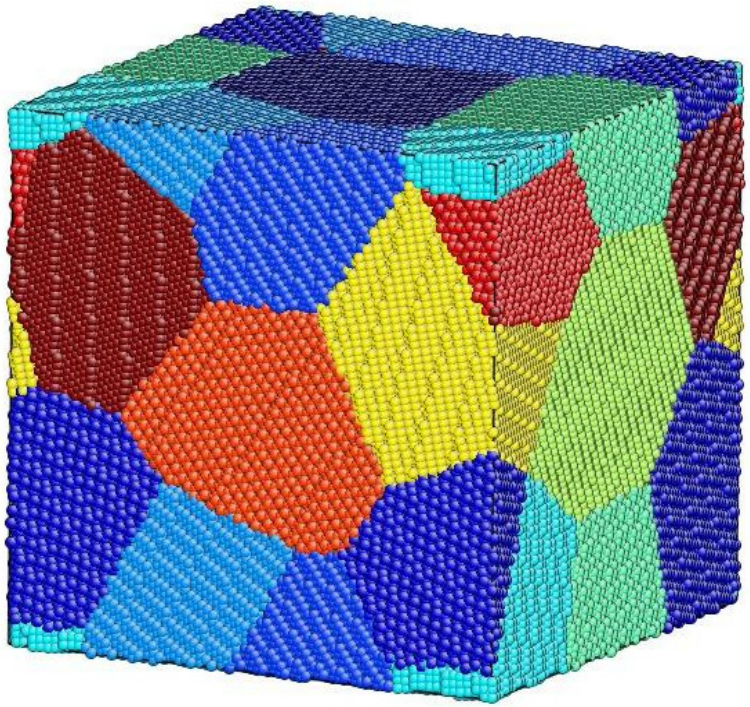
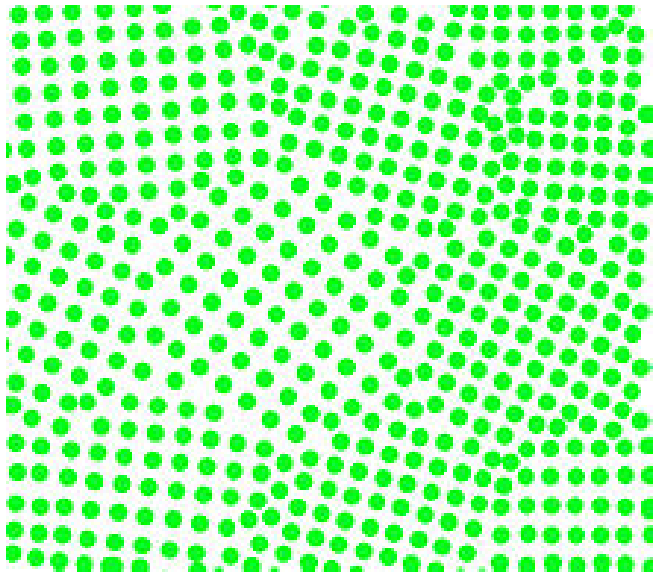


Figure: Screw dislocation motion during in situ TEM straining. The dislocation$A$ cross-slip from (321) plane to (110) plane. Source: Philippe Castany, M. Besse, Thierry Gloriant. Scripta Materialia, 2012, 66 (6), pp.371

Grain-size reduction

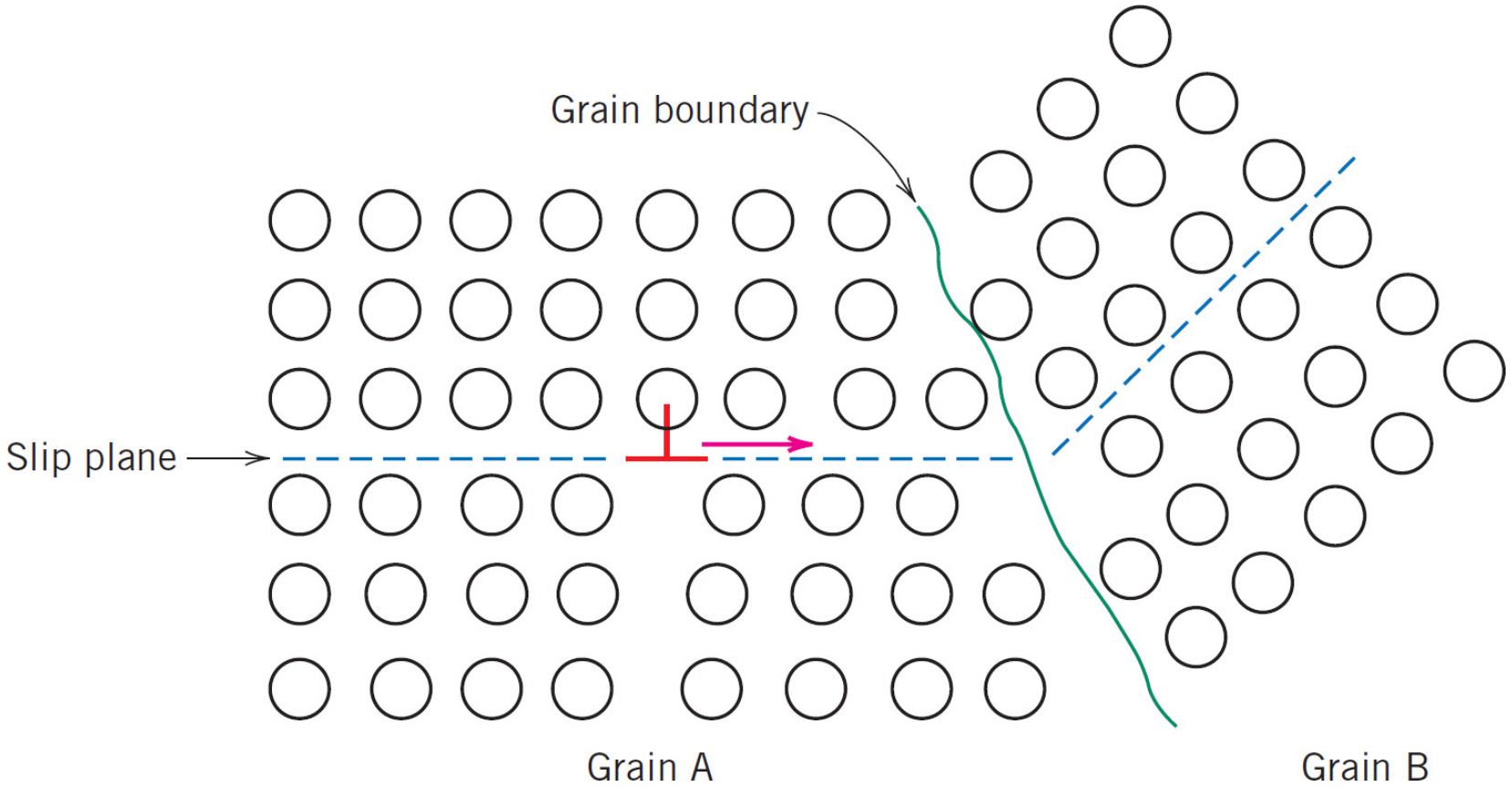
- The grain size (average grain diameter), in a polycrystalline metal influences the mechanical properties.
- Nearby grains normally have different crystallographic orientations.





Grain-size reduction

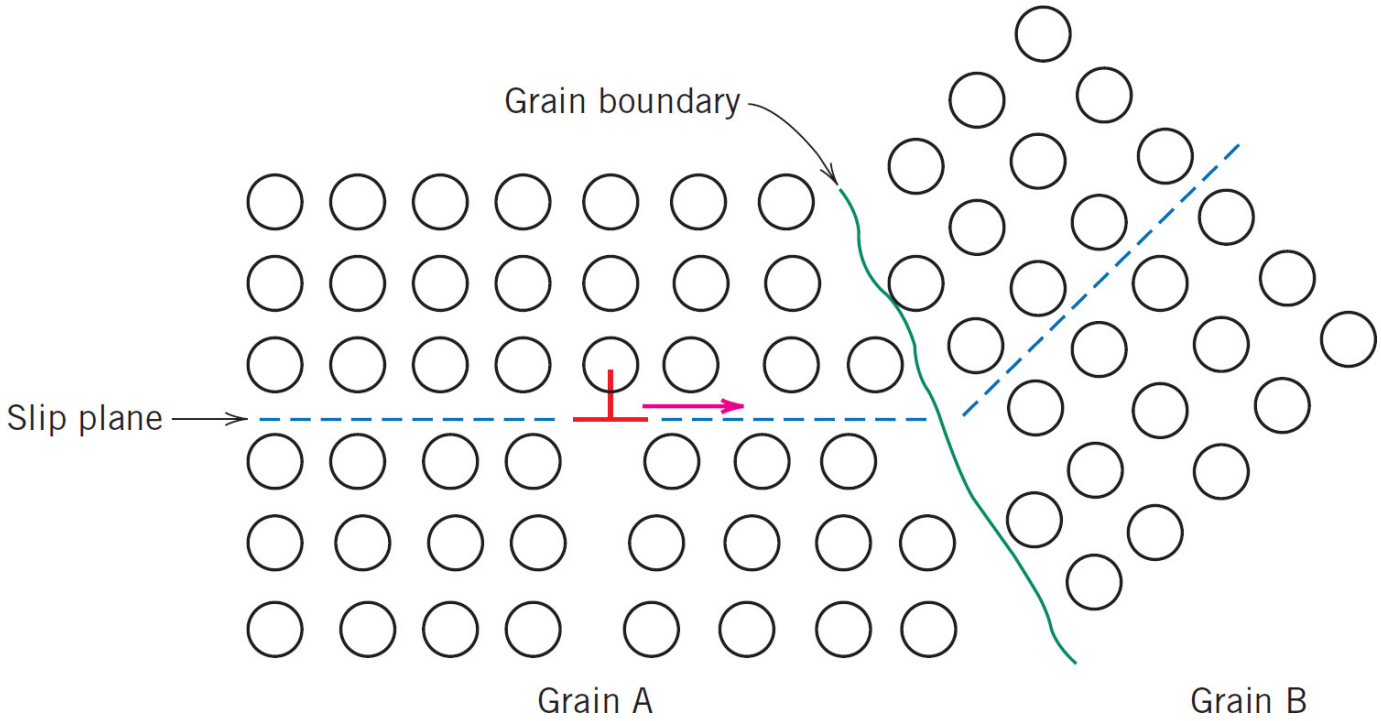
During plastic deformation or slip, dislocations must move across the common grain boundary, for example from grain A to grain B.



Grain-size reduction

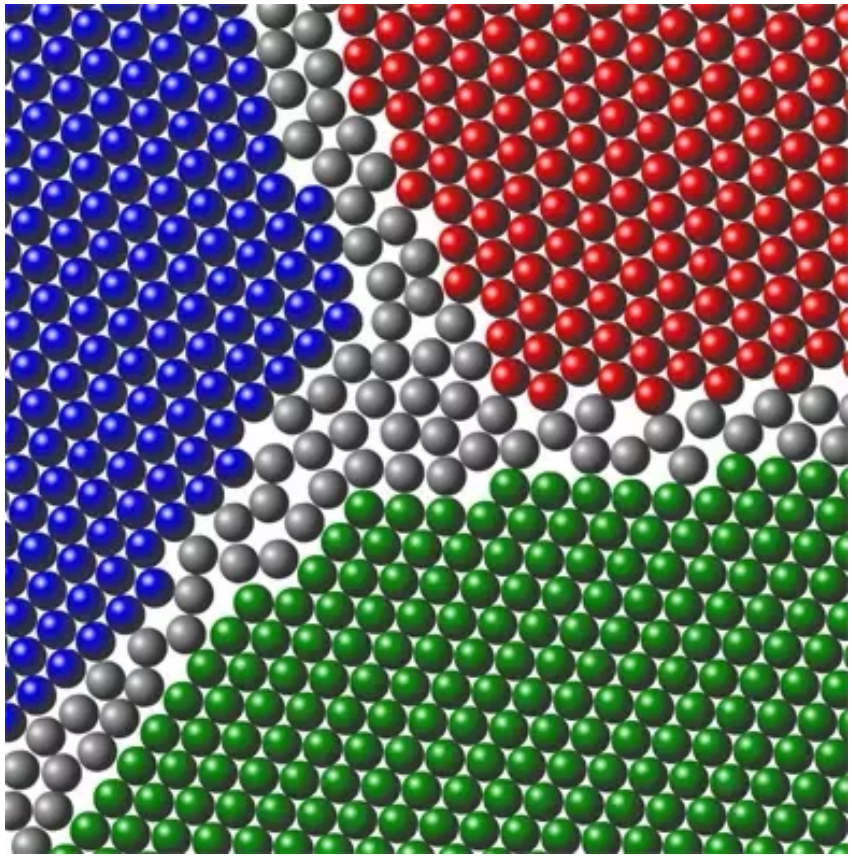
The grain boundary is a barrier for a dislocation:

- The two grains have different orientations.
- A dislocation passing into grain B will have to change its direction of motion.
- This becomes more difficult if the crystallographic misorientation increases.



Grain-size reduction

- The grain boundary is a barrier for a dislocation.
- The atomic disorder within a grain boundary region will be a discontinuity of slip planes from one grain into the other.



Grain-size reduction

- A fine-grained material (with small grains) is harder and stronger than one that is coarse grained (large grains).
- This is because a fine grained material has a greater total grain boundary area.
- For many materials, the yield strength σ_y varies with grain size according to the **Hall-Petch** equation.

Hall-Petch equation

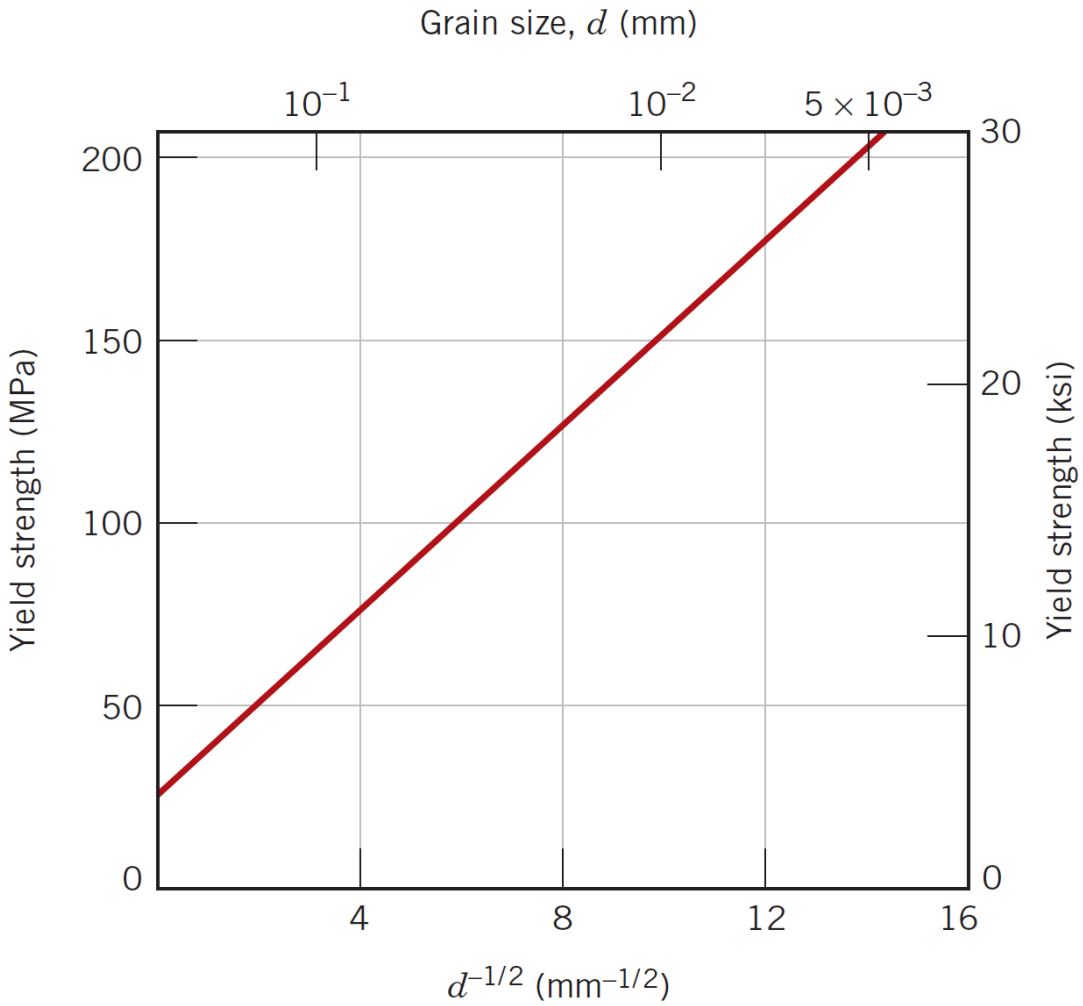
$$\sigma_y = \sigma_0 + \left(\frac{k}{\sqrt{d}}\right)$$

d Average grain diameter

σ_0, k Constants for the material



Grain-size reduction



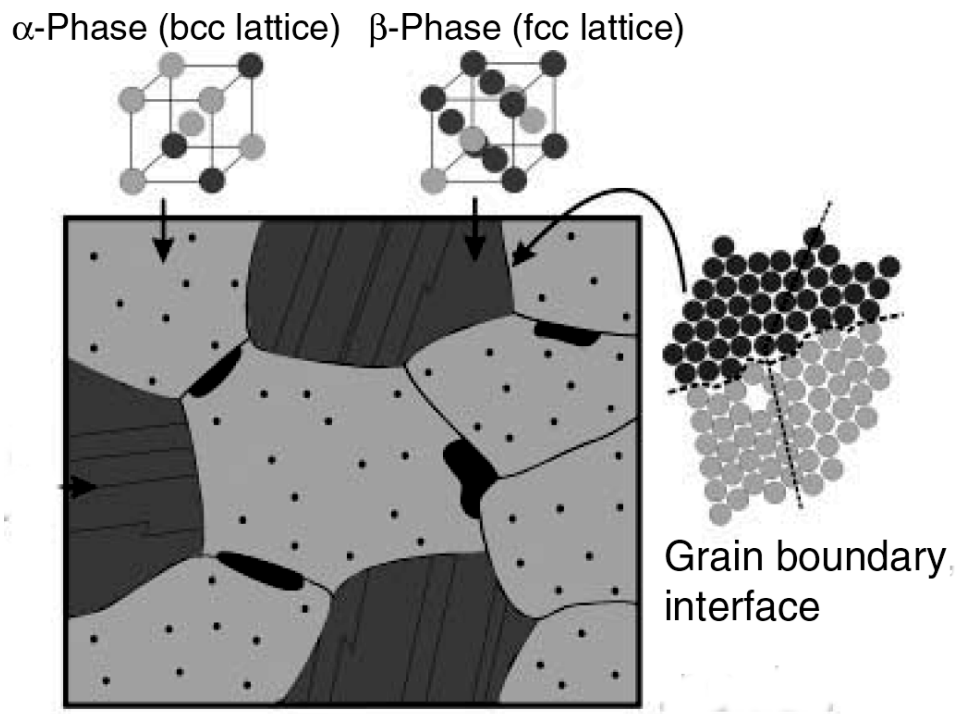
$$\sigma_y = \sigma_0 + \left(\frac{k}{\sqrt{d}}\right)$$

Hall-petch equation is **Not valid** for very coarse and extremely fine-grained materials.

Figure: Effect of grain size on yield strength of a 70 Cu–30 Zn brass alloy.

Grain-size reduction

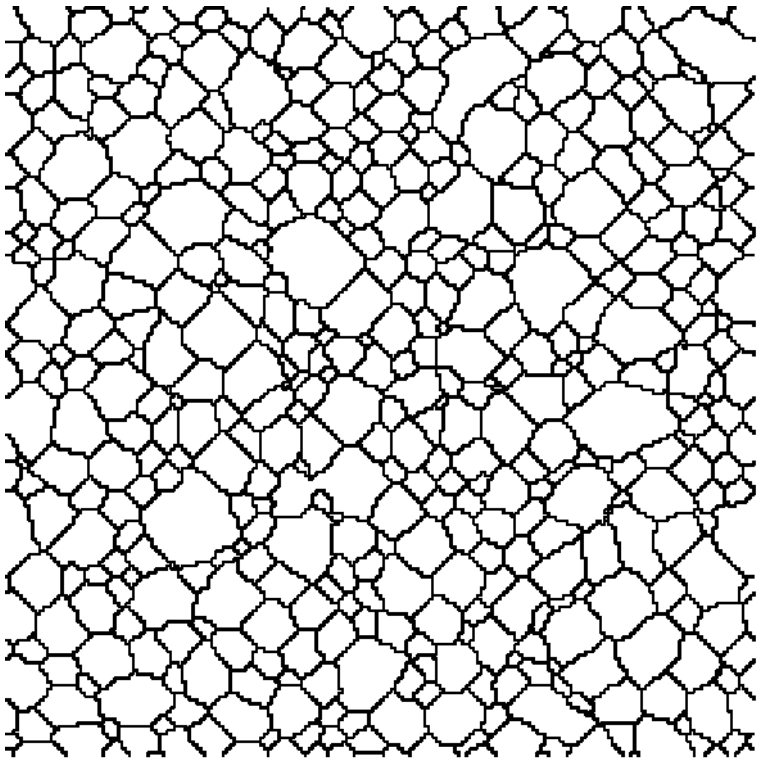
- Grain size may be changed during solidification, plastic deformation and heat treatment.
- Grain size reduction improves not only strength, but also the toughness of many alloys.
- Grain boundaries between two different phases can also be a barrier to for dislocations important in the strengthening of multiphase alloys.



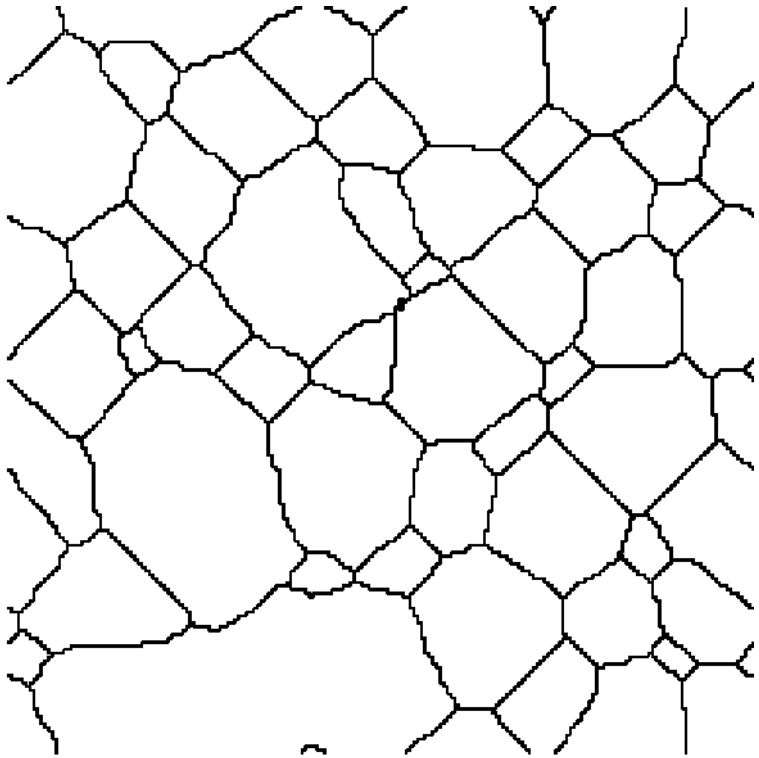
Grain-size reduction

Question

If the folowing samples are of the same alloy and at the same magnification, which sample has the higher yield strength?



A.



B.

Solid solution strengthening

- Alloying with impurity atoms is a method for increasing strength and hardness of pure metals.
- The impurity atoms can form either substitutional or interstitial **solid solution**.
- Therefore, this is called **solid-solution** strengthening.
- High-purity metals are softer and weaker than their alloys.

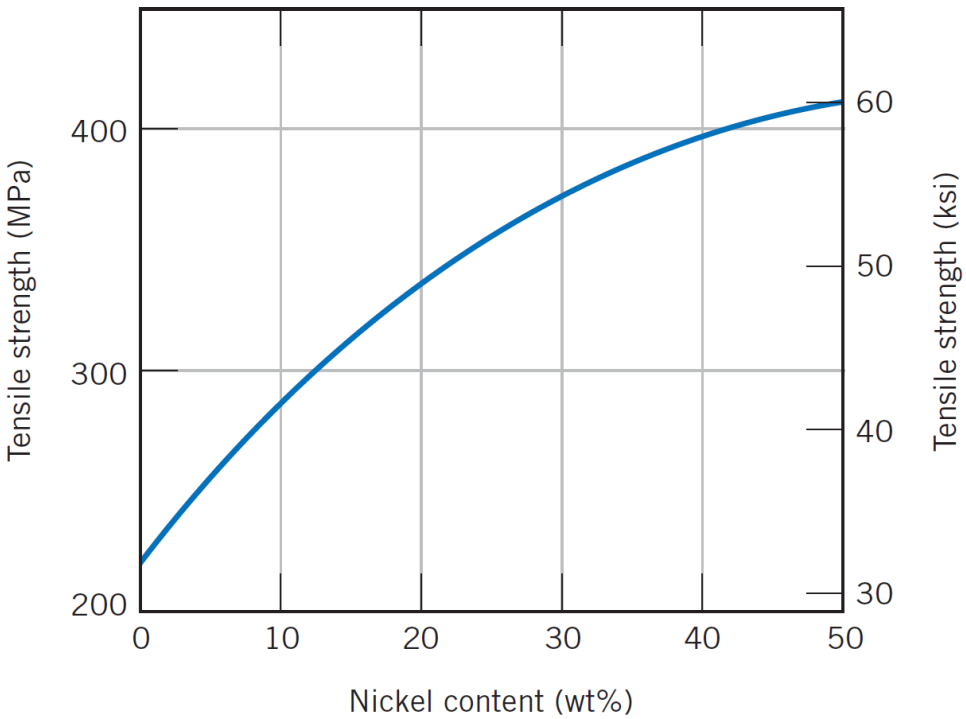


Figure: Variation of tensile strength, with Ni content in Cu-Ni alloys.

Solid solution strengthening

The yield strength increases while ductility decreases with impurity content.

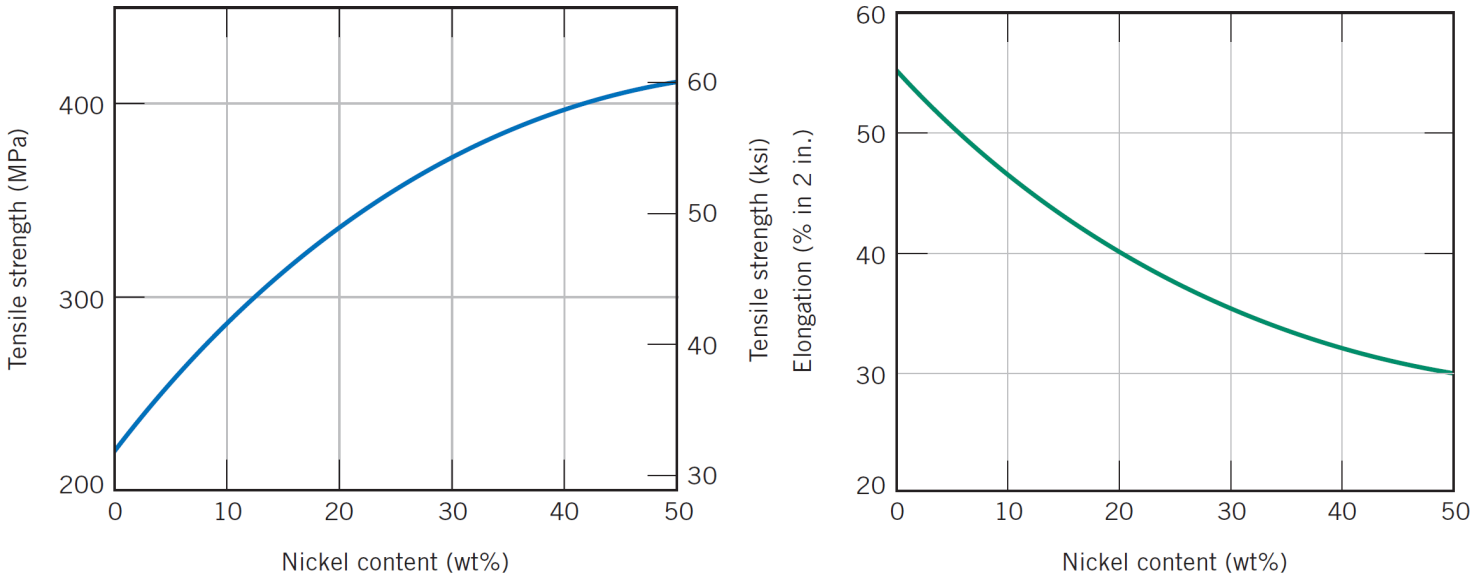
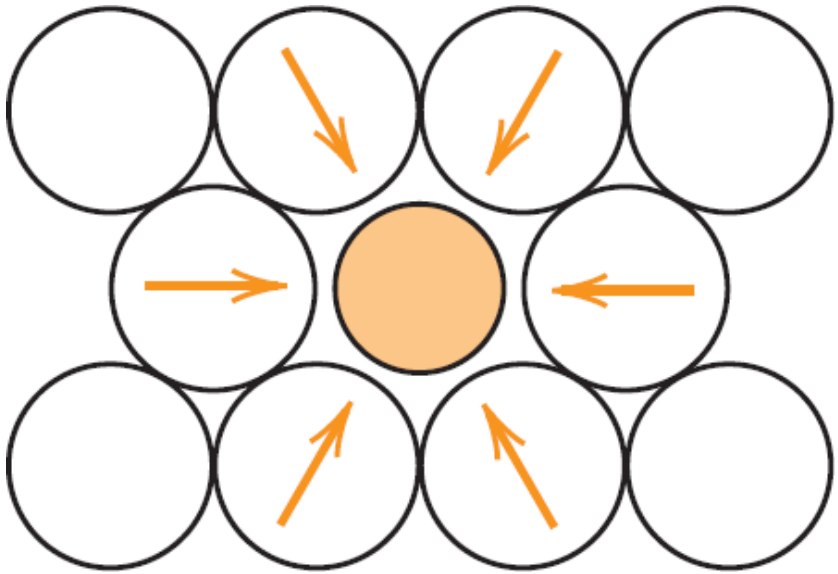


Figure: Variation of the yield strength and ductility, with the nickel content in Cu-Ni alloys.

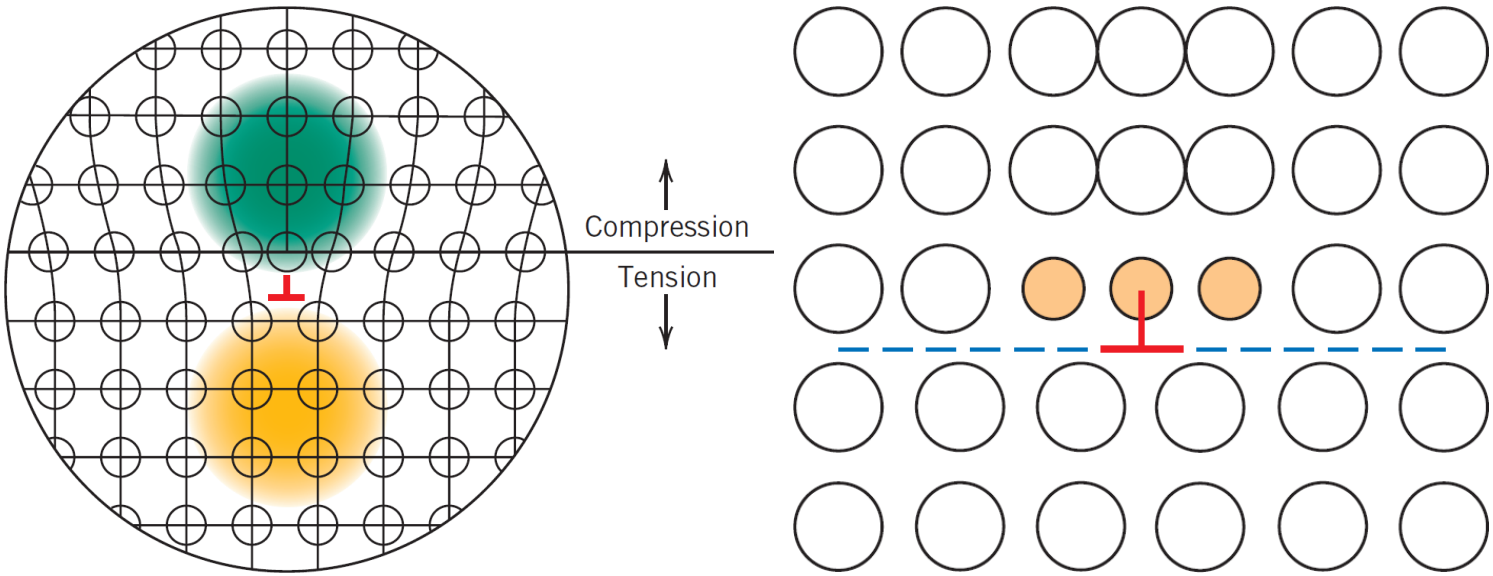
Solid solution strengthening

- Solid solution impurity atoms cause lattice strains on the surrounding host atoms.
- Lattice strain field interacts with dislocations.
- Therefore, dislocation movement is restricted.
- A smaller impurity atom exerts tensile strains on the surrounding crystal lattice.



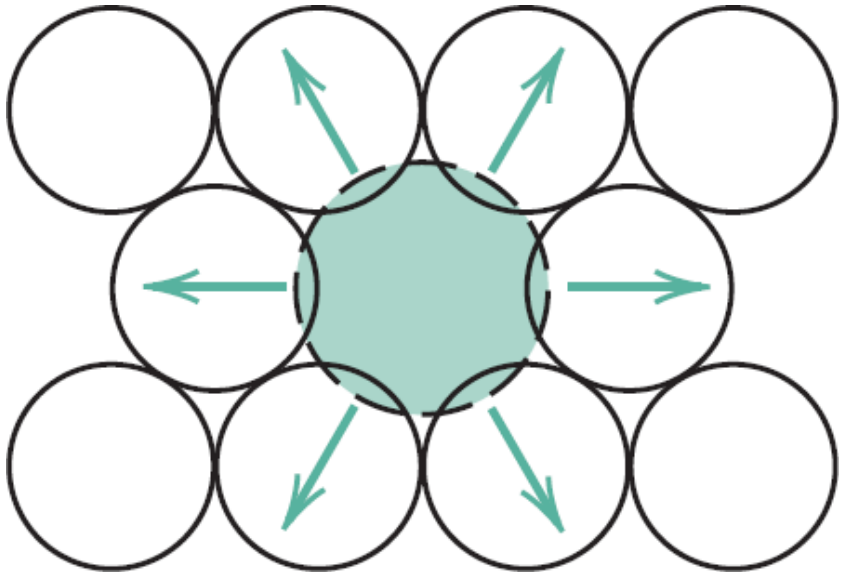
Solid solution strengthening

- Smaller impurity atoms may diffuse to dislocation sites to reduce the overall compressive strain energy around a dislocation.
- This location would be near the dislocation line and **above** the slip plane as shown.



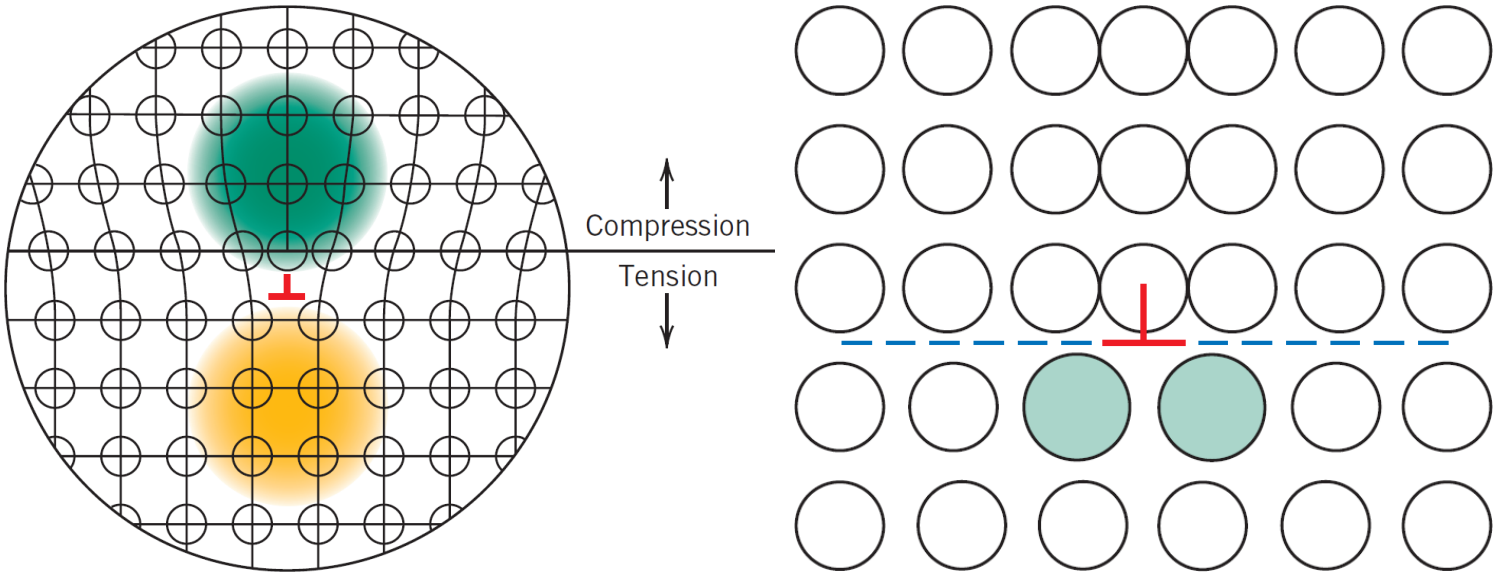
Solid solution strengthening

A larger impurity atom exerts compressive strains on the surrounding crystal lattice.



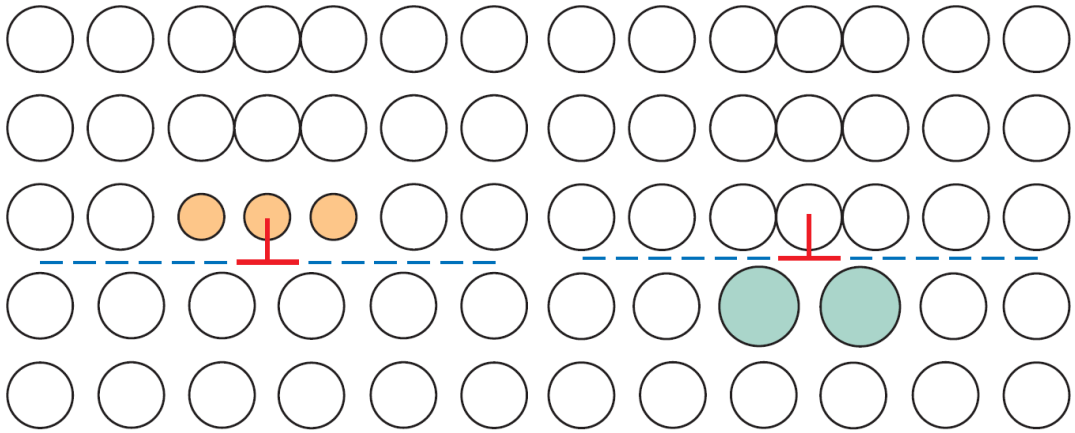
Solid solution strengthening

- Larger impurity atoms may diffuse to dislocation sites to reduce the overall **tensile** strain energy around a dislocation.
- This location would be near the dislocation line and **below** the slip plane as shown.



Solid solution strengthening

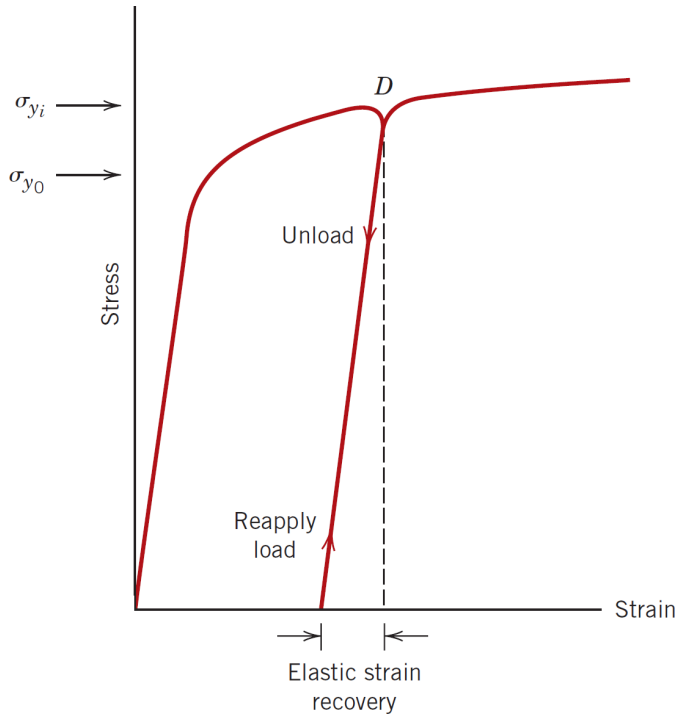
- In terms of energy, the overall strain energy must increase if a dislocation is moved away (therefore naturally discouraged).
- Even during dislocation motion, the lattice strains between impurity atoms and dislocations interact.
- A greater applied stress is necessary to first start and then continue plastic deformation for solid-solution alloys, compared to pure metals.
- Therefore, strength and hardness are increased.

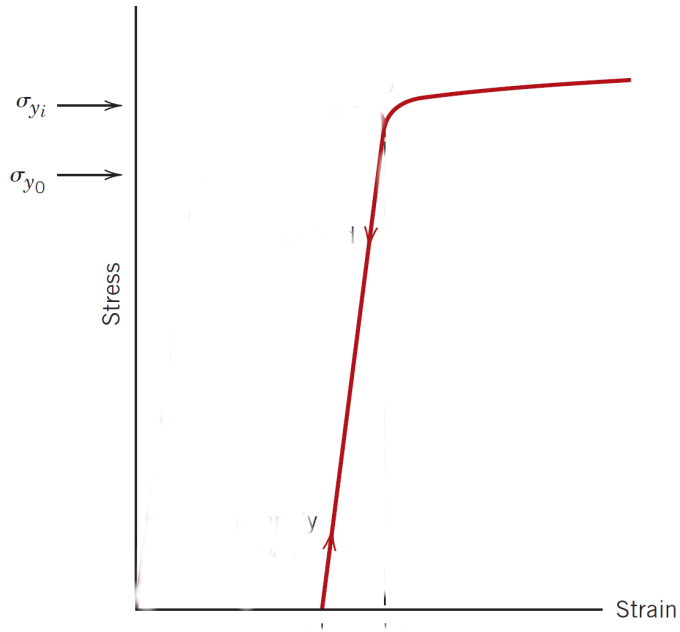


Strengthening methods

Strain hardening

- Strain hardening: a ductile metal becomes harder and stronger as it is plastically deformed.
- Sometimes strain hardening is also called **work hardening** or **cold working**.
- Most metals strain-harden at room temperature.





It is sometimes convenient to express the degree of plastic deformation as percent cold work rather than as strain.

Percent cold work

$$\%\mathrm{CW} = \left(\frac{A_0 - A_d}{A_0}\right) \times 100$$

A_0 The original cross sectional area undergoing deformation

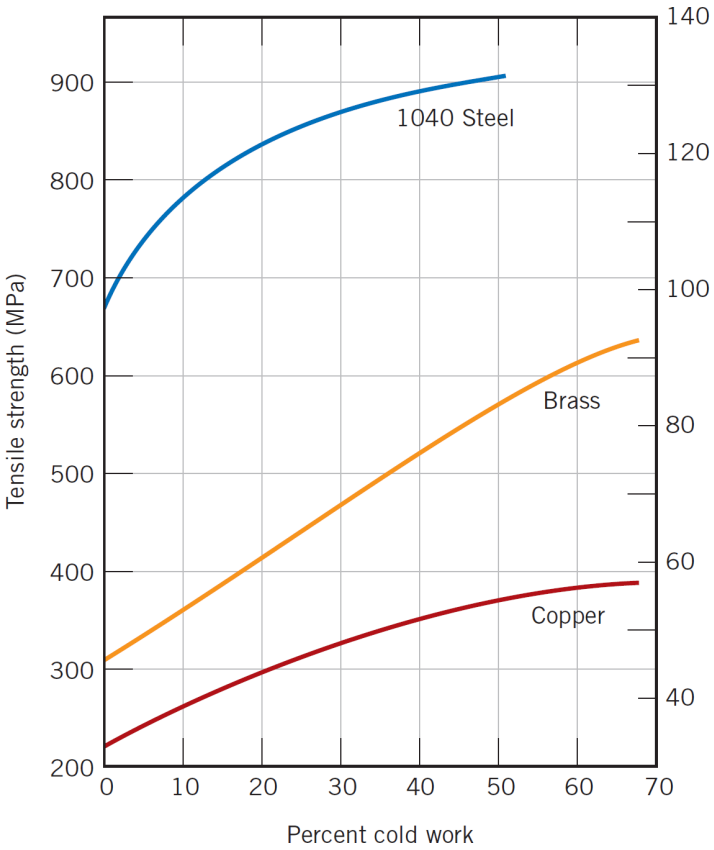
A_d Area after deformation

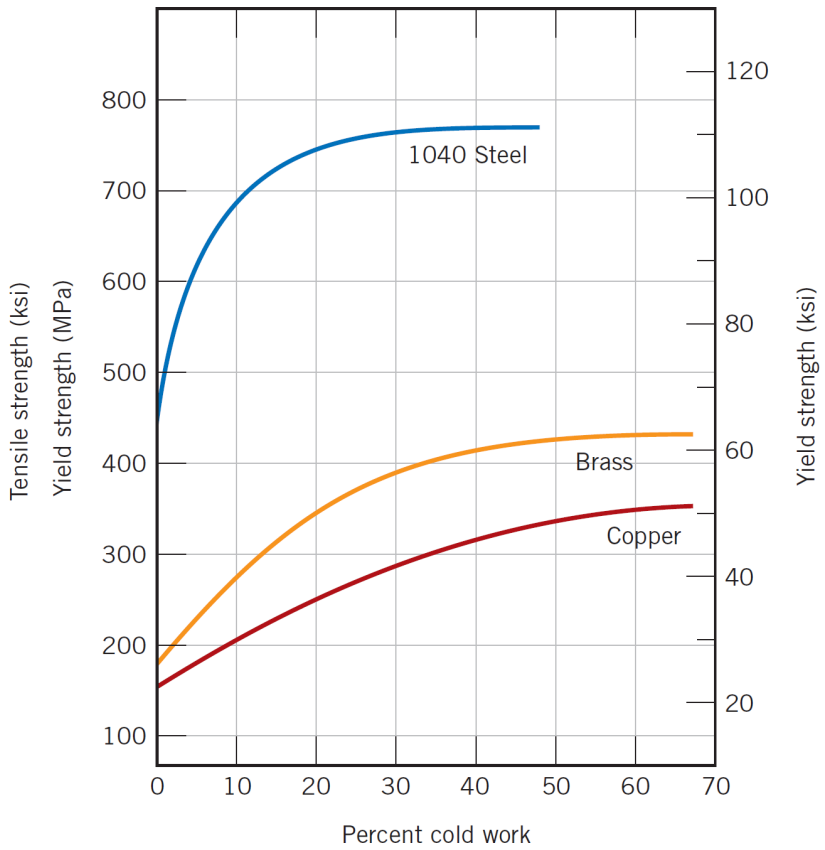


Strengthening methods

Strain hardening

- Most materials can be strain-hardened to increase yield, strength and tensile strength.
- The graphs show how steel, brass, and copper increase in yield and tensile strength with increasing cold work.

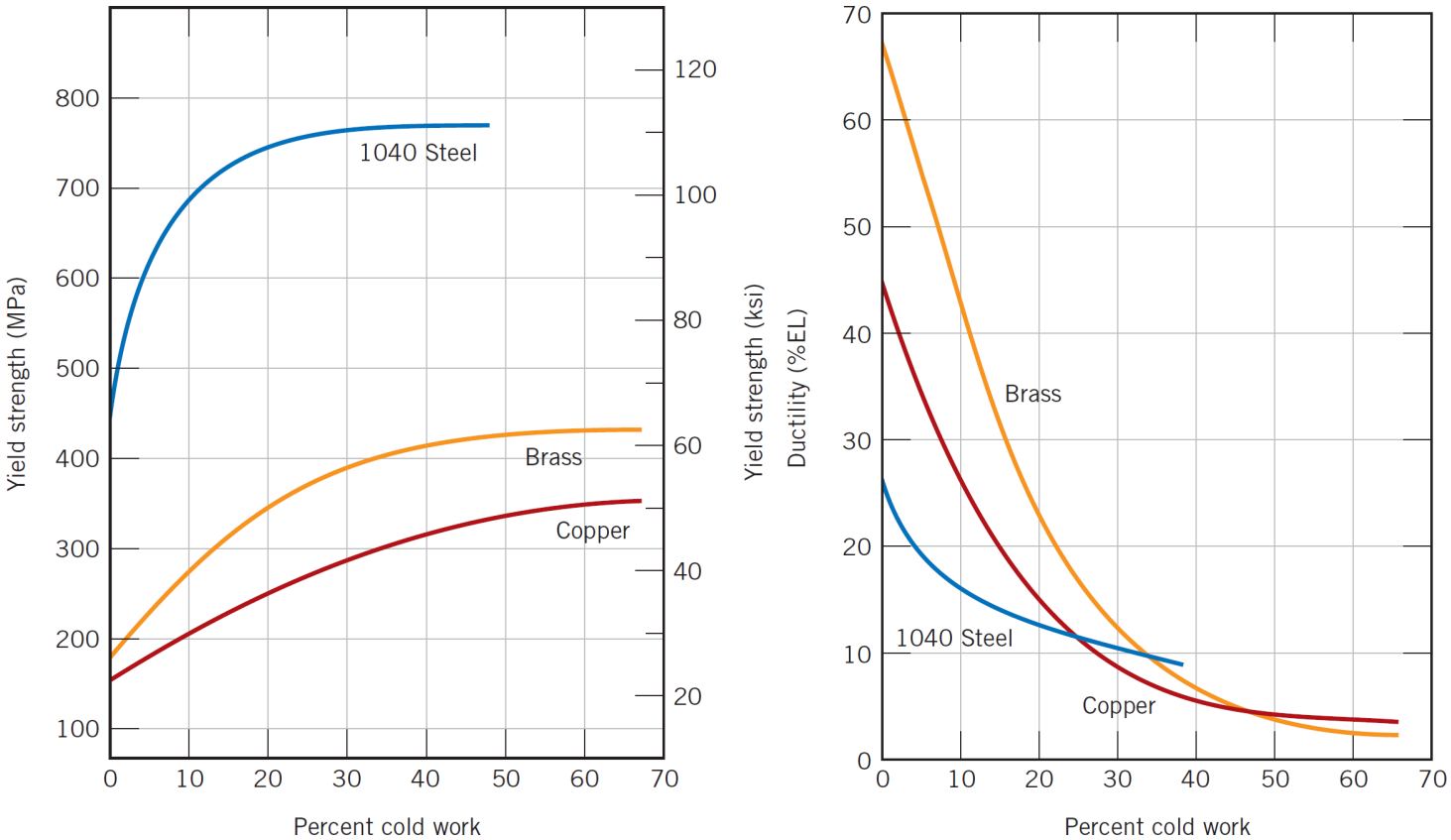




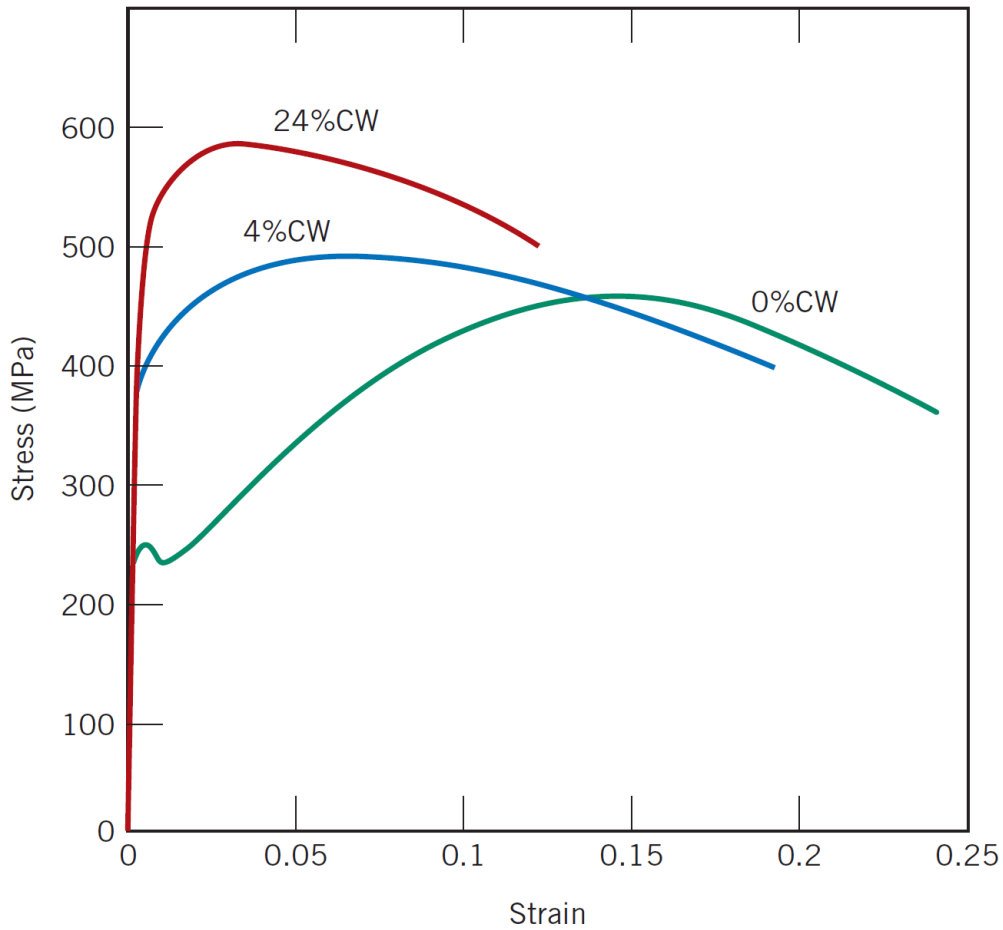
Strengthening methods

Strain hardening

- The enhancement of hardness and strength comes at a price.
- Ductility (percent elongation), undergoes a reduction with increasing percent cold work for the same three alloys.



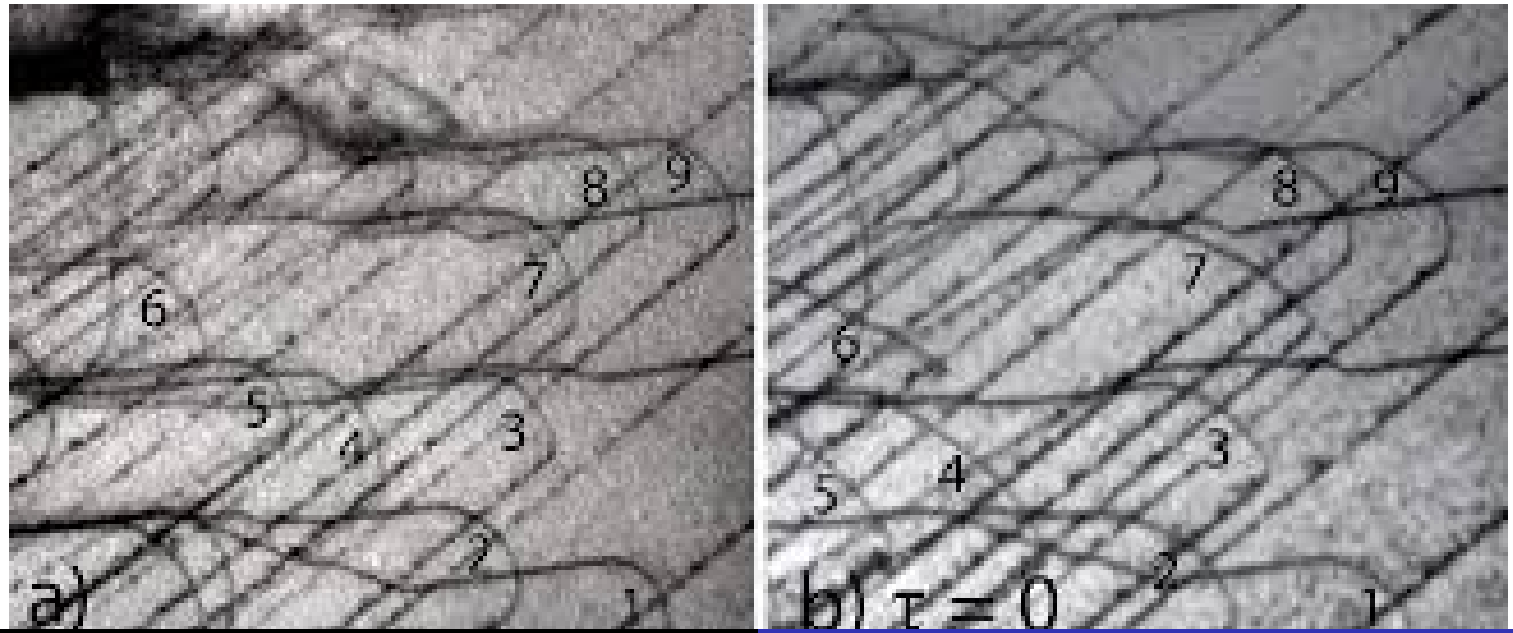
The effect of cold work on the stress–strain behavior of a low-carbon steel is shown below



Strengthening methods

Strain hardening

- The strain-hardening can be explained with dislocation–dislocation strain field interactions.
- The dislocation density in a metal increases with deformation or cold work.
- Dislocations multiply and sometimes new dislocations are formed during cold work.
- The average distance of separation between dislocations decreases and they come closer together.



- On the average, dislocation–dislocation strain interactions are repulsive.
- The result is that dislocation motion is stopped by the presence of other dislocations.
- As the dislocation density increases, this resistance to dislocation motion by other dislocations becomes more and more.
- Therefore, the stress necessary to deform a metal increases with increasing cold work.



- Strain hardening is used commercially to improve the mechanical properties of metals during fabrication processes.
- The effects of strain hardening may be removed by an annealing heat treatment

